

<https://doi.org/10.1038/s41524-025-01704-4>

Materials design with target-oriented Bayesian optimization



Yuan Tian¹, Tongtong Li², Jianbo Pang³, Yumei Zhou³, Dezhen Xue³✉, Xiangdong Ding³✉ & Turab Lookman^{3,4}✉

Materials design using Bayesian optimization (BO) typically focuses on optimizing materials properties by estimating the maxima/minima of unknown functions. However, materials often possess good properties at specific values or show effective response under certain conditions. We propose a target-oriented BO to efficiently suggest materials with target-specific properties. The method samples potential candidates by allowing their properties to approach the target value from either above or below, minimizing experimental iterations. We compare the performance of target-oriented BO with that of other BO methods on synthetic functions and materials databases. The average results from hundreds of repeated trials demonstrate target-oriented BO requires fewer experimental iterations to reach the same target, especially when the training dataset is small. We further employ the method to discover a thermally-responsive shape memory alloy $\text{Ti}_{0.20}\text{Ni}_{0.36}\text{Cu}_{0.12}\text{Hf}_{0.24}\text{Zr}_{0.08}$ with a transformation temperature difference of only 2.66 °C (0.58% of the range) from the target temperature in 3 experimental iterations. Our method provides a solution tailored for optimizing target-specific properties, facilitating the accelerated development of materials with predefined properties.

Bayesian optimization (BO) is widely used in materials design within an active learning loop or Bayesian framework to minimize the number of experiments for finding desired properties where training data is limited^{1–4}. It employs acquisition functions to identify the most informative candidates by ranking unknown options in the materials parameter space, thereby guiding the next experiment in the black-box optimization process^{5–8}. For example, pure exploration⁹ has been employed to rapidly construct phase diagrams^{10–12}, learn potential energy surfaces¹³, and build machine-learned potentials¹⁴. Since the introduction of the concept of improvement from the best so far within a Bayesian framework, variants such as Efficient Global Optimization (EGO)¹⁵, Knowledge Gradient and Upper Confidence Bound^{16,17}, have been widely applied to design problems in materials discovery^{18–20} to find shape memory alloys with very small hysteresis²¹, ferroelectrics with large electrostrains²² and high entropy alloys with high hardness²³. Recently, the acquisition function EI in EGO is further modified to tackle various specific material issues. Variants of EI evaluating the improvement over the observed convex hull rather than total energy minimization are demonstrated to identify the ground-state energy line of a configurational space^{24,25}. Multi-objective Acquisition Functions(MOAF) are proposed to seek a Pareto-front solution for acquisition function values, significantly improving the robustness of acquisition functions²⁶. Current

optimization strategies predominantly focus on finding the maxima or minima of the objective functions or material properties^{10,27}.

However, in many materials applications (Fig. 1), the values of target-specific properties are of greater interest than merely finding their optimal values, as materials often show exceptional performance at specific property values or exhibit effective response under certain conditions, which may not necessarily involve maximizing or minimizing a property^{28,29}. For example, catalysts for hydrogen evolution reactions have enhanced activities when the free energies approach zero³⁰. Similarly, photovoltaic materials^{31,32} exhibit high energy absorption within a targeted range of the band gap energy, endovascular stent materials deform at a body transition temperature close to 37 °C and thermoelastic shape memory alloys show small hysteresis under elastic compatibility conditions when the middle eigenvalue, λ_2 , of the strain tensor equals 1³³. Thus, there's a need to pinpoint materials with predefined specific properties.

Intuitively, materials with target-specific properties can be designed based on predicted outcomes from machine learning models, such as ensemble learning model³⁰, that rely on model accuracy. To address this problem, Chitturi et al. employ a Gaussian process model to predict a subset of the design space that satisfies the experimental goal, followed by an uncertainty-aware acquisition function to guide the subsequent

¹Materials Genome Institute, Shanghai University, Shanghai, 200444, China. ²School of Materials Science and Engineering, Zhejiang Sci-Tech University, Hangzhou, 310018, China. ³State Key Laboratory for Mechanical Behavior of Materials, Xi'an Jiaotong University, Xi'an, 710049, China. ⁴AiMaterials Research, Santa Fe, New Mexico, 87501, USA. ✉e-mail: xuedezhen@xjtu.edu.cn; dingxd@mail.xjtu.edu.cn; turablookman@gmail.com

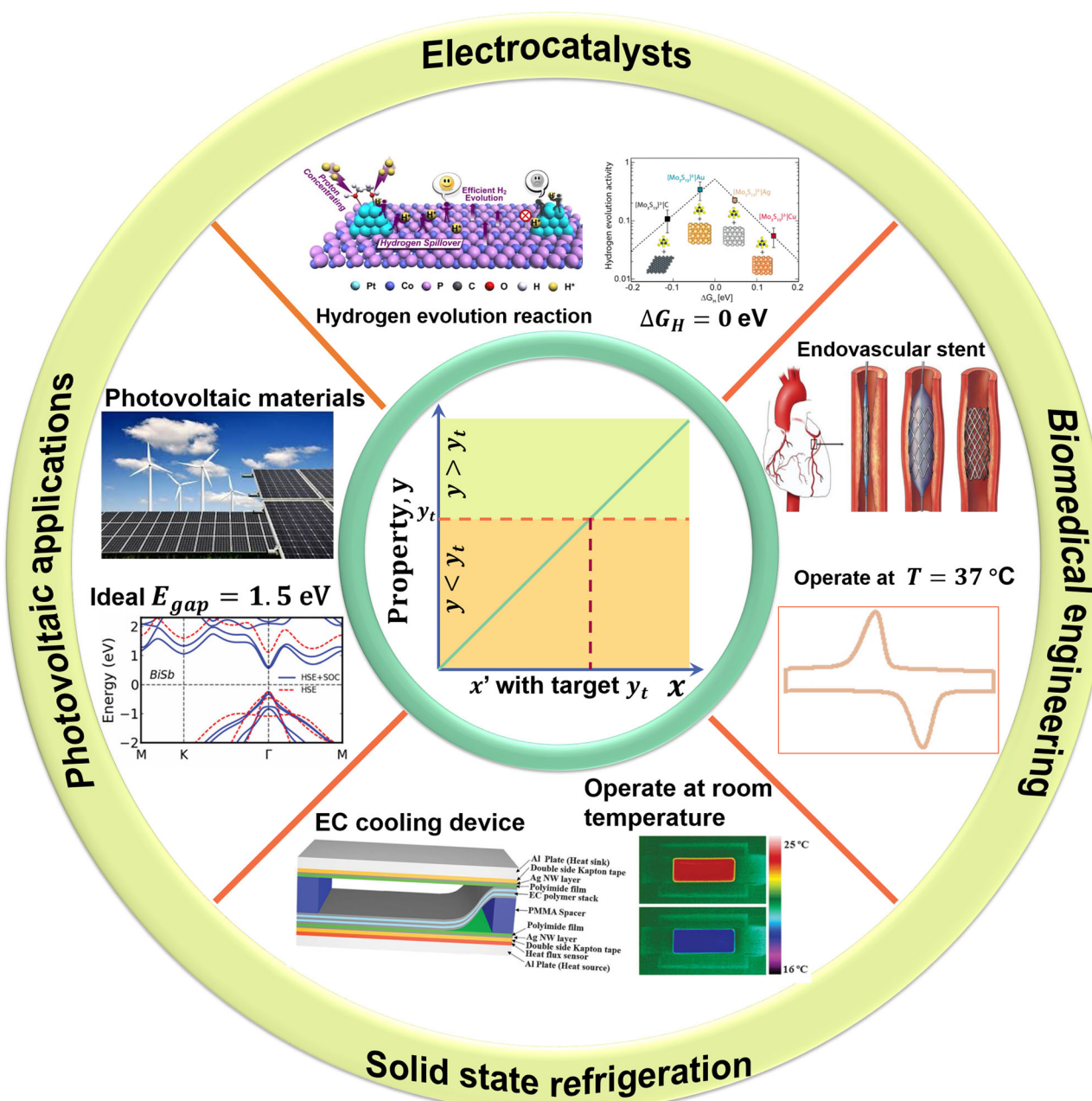


Fig. 1 | Applications of materials with specific properties. Materials with specific properties show applications in the fields of photovoltaic, electrocatalysts, biomedical engineering and solid-state refrigeration.

measurement³⁴. This approach performs well when the model's prediction uncertainty is low, but may face the risk of high predictive uncertainty, leading to potential candidates falling outside the predefined subset. Although few studies focus on minimizing experiments in the search for target-specific values, some have proposed strategies that transform the problem to a minimizing task by redefining the property as the deviation from the target^{30,34}. For example, to predict 2D hexagonal binary compounds with a band gap energy (E_{gap}) equal or close to 1.5 eV³², Ma et al. treat $|E_{\text{gap}} - 1.5|$ as the objective function to be minimized within a BO loop using EGO. However, as its acquisition function EI calculates the expected improvement from the current best value to infinity instead of zero, this results in suggesting experiments that are less than optimal. Therefore, there arises a necessity to develop a new strategy that fully assesses the potential information with considering uncertainties from all candidates in the design space to accelerate the discovery of materials with target-specific property values.

In this work, we develop a target-oriented Bayesian optimization method t-EGO for a target-specific property, which employs the acquisition function t-EI to sample by tracking the difference from the desired property with its associated uncertainty. We first test the efficiency of the method on several mathematical functions of varying complexities and compare it to EGO, constrained EGO, MOAF, Pure exploitation (PureExp) and others. We next validate t-EGO on searching a hydrogen evolution reaction (HER) catalyst with a target hydrogen adsorption free energy of zero using the DFT calculation database for two-dimensional layered MA_2Z_4 materials³⁰. The statistical results from hundreds of repeated trials indicate that t-EGO demonstrates superior performance over other BO methods, especially requiring approximately 1 to 2 times fewer experimental iterations than EGO/MOAF strategy to reach the same target. Finally, t-EGO is employed to develop thermally-responsive shape memory alloys (SMAs) with a phase transformation temperature of 440 °C for use as a thermostatic valve material that automatically regulates the main steam temperature of

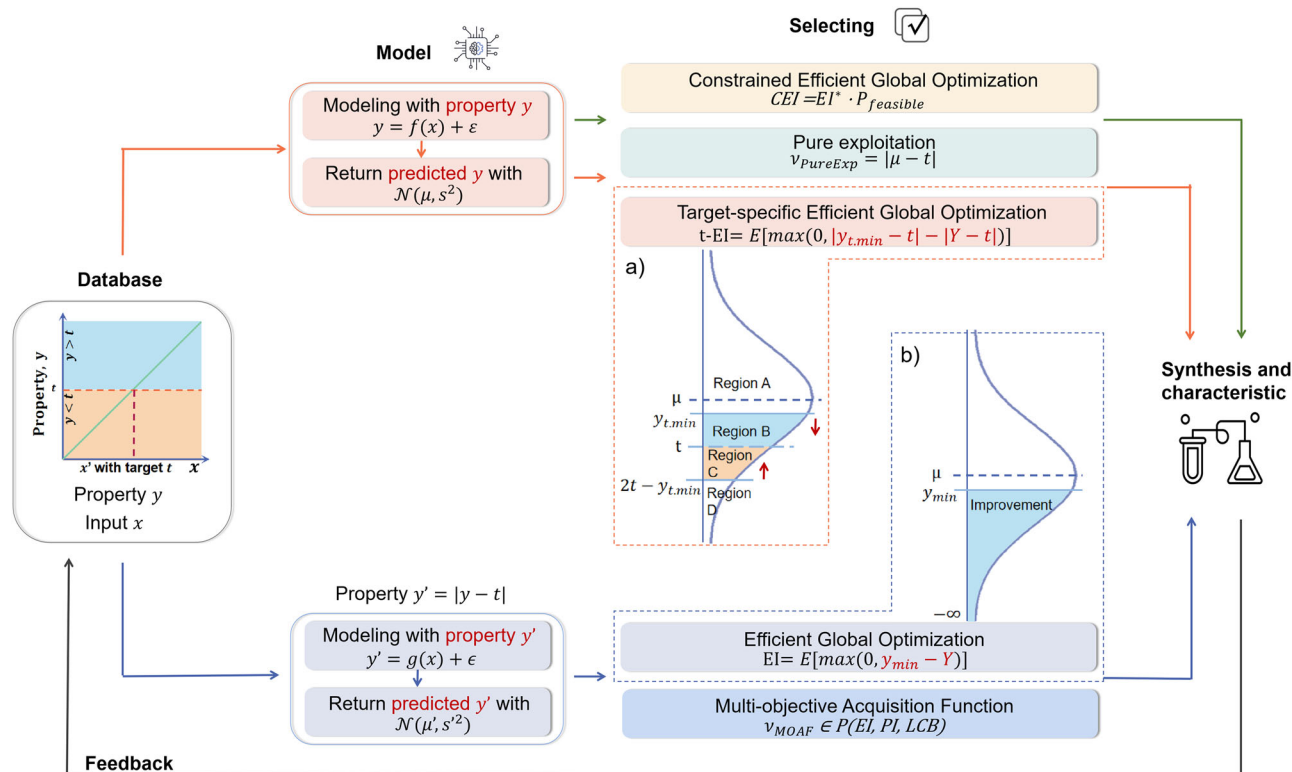


Fig. 2 | Targeted-oriented Bayesian optimization t-EGO compared with Pure-Exp, EGO, CEGO and MOAF. t-EGO for $y_{t,min} > t$ and EGO are shown in (a, b). **a** The region under the normal distribution curve is divided into Regions A, B, C and D. **b** The solid horizontal line denoted as $y_{t,min}$ represents the measurement value in

the training set closest to the targeted property value (t). The horizontal line at $2t - y_{t,min}$ represents the symmetric position of the line at $y_{t,min}$ with respect to t . **b** The solid horizontal line at y_{min} indicates the current minima.

turbines. We synthesized the SMA $\text{Ti}_{0.20}\text{Ni}_{0.36}\text{Cu}_{0.12}\text{Hf}_{0.24}\text{Zr}_{0.08}$ with a temperature of 437.34 °C within 3 experimental iterations, achieving a temperature difference of only 2.66 °C from the target. The success with which we find the SMA with target-specific transformation temperature indicates that t-EGO is particularly suited for finding materials with desired properties or satisfying specific requirements.

Results

Target-oriented Bayesian optimization

We propose a novel target-oriented Bayesian optimization method, t-EGO, to identify materials with target-specific properties using the minimal number of experiments, as illustrated in Fig. 2. To assess the effectiveness of our method, we also conducted a comparative analysis of multiple approaches, including Constrained EGO (CEGO)^{35,36}, pure exploitation (PureExp) as well as the objective-reformulated method. In the objective-reformulated method, the objective y is reformulated as the distance from the target value t , i.e., $y' = |y - t|$. The target-specific property is then achieved by minimizing y' using extremum optimization algorithms, such as EI and Multi-Objective Acquisition Functions (MOAF)²⁶. The overall design loops of all methods are presented in Fig. 2.

In the *modeling* process, t-EGO, PureExp and CEGO strategies use unprocessed data with property y as inputs for modeling. In contrast, the EGO and MOAF strategies construct models based on training data y processed by $|y - t|$, where t is the target-specific value. In the *selecting* process, PureExp recommends candidates solely in terms of predicted values from the model, without considering prediction uncertainties. t-EGO, EGO and CEGO involve acquisition functions based on expected improvements accounting for prediction uncertainties, where the property value at x is predicted as Y , a random variable that follows a normal distribution ($\mathcal{N}(\mu, s^2)$) with mean μ and variance s^2 . Different from the Expected Improvement (EI) in EGO, both target-specific Expected

Improvement (t-EI) in t-EGO and constrained Expected Improvement (CEI) in CEGO, incorporate the distance to the target-specific value t , thereby constraining the distribution of the predicted values. Here, we will provide a detailed derivation of t-EI from EI, while the formulas for other algorithms are listed in Supplementary Table 1. The schematic diagrams for t-EI and EI are shown in Fig. 2a, b.

Expected Improvement in EGO

Given the best (minimum in a minimization problem) property value, y_{min} , obtained so far, the improvement over y_{min} at a point x can be expressed by $I = \max(y_{min} - Y, 0)$. Then the expected improvement (EI) is given by^{1,21,37}

$$EI = E[\max(0, y_{min} - Y)] \quad (1)$$

where $y_{min} = \min(y_1, \dots, y_n)$ is the minimum value of the n experimental measurements in the training set. The random variable Y can be reparameterized as $Y = \mu + sz$, where z follows a standard normal distribution. Thus, EI can be evaluated using its definition:

$$\begin{aligned} EI &= \int_{-\infty}^{\infty} I(x)\phi(z)dz \\ &= s \int_{-\infty}^{y_{min}-\mu} \left(\frac{y_{min}-\mu}{s} - z\right)\phi(z)dz \\ &= (y_{min} - \mu)\Phi\left(\frac{y_{min}-\mu}{s}\right) + s\phi\left(\frac{y_{min}-\mu}{s}\right) \end{aligned} \quad (2)$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the probability density and cumulative distribution functions of the normal distribution, respectively.

Target-specific Expected Improvement in t-EGO

The goal of the acquisition function t-EI is to sample the candidate so that the next measured property value is closer to the target value than the current property value closest to the target. As shown in Fig. 2a, suppose that

a fixed value t represents the target property, and $y_{t,min}$ and $2t - y_{t,min}$ represent the property value in the training dataset closest to the target property value t and symmetric position of $y_{t,min}$ with respect to t . Define $Dis_{min} = \min(|y_1 - t|, |y_2 - t|, \dots, |y_n - t|) = |y_{t,min} - t|$ as the minimum difference between t and the property values ($y_1, y_2, \dots, y_n$) measured so far. Also, $|Y - t|$ indicates the difference value between the predicted property value Y of the unknown material x and the targeted property value t . If $|Y - t| < |y_{t,min} - t|$, then x given by t-EI will bring the improvement in $|y_{t,min} - t| - |Y - t|$. Otherwise, x will provide no improvement on the current state. Hence, we define t-EI as:

$$t - EI = E[\max(0, |y_{t,min} - t| - |Y - t|)] \quad (3)$$

In Fig. 2a, predicted Y of a unlabeled material that lies in Region B between $y_{t,min}$ and t , as well as Region C between t and $2t - y_{t,min}$ holds a $|Y - t|$ smaller than Dis_{min} . It thus represents the property value closer to the preset target value and has the potential of improvement based on the current data according to Equation (3). In Region A and D, the difference between t and predicted Y of unlabeled materials is greater than Dis_{min} , and a data in this region will have no improvement on the current situation. Therefore, regions B and C are considered as the feasible regions with the intervals of $[y_{t,min}, t]$ for region B and $[t, 2t - y_{t,min}]$ for region C, while regions A and D are deemed infeasible. The exploration is conducted within the feasible regions, guiding the candidate properties toward the target value from either direction, above or below, as indicated by the red arrows in Fig. 2a.

Thus, the distribution of the predicted value Y in t-EI is constrained within the interval of $[y_{t,min}, t]$ and $[t, 2t - y_{t,min}]$, instead of $[-\infty, y_{min}]$ or $[y_{max}, \infty]$ in EI, as shown in Fig. 2a and b. Equation (3) can be further rewritten by calculating the conditional expectation:

$$\begin{aligned} t - EI &= EI^B + EI^C \\ &= E[Y_{t,min} - Y | y_{t,min} > Y > t] \\ &\quad + E[Y - (2t - y_{t,min}) | t > Y > 2t - y_{t,min}] \end{aligned} \quad (4)$$

where

$$\begin{aligned} EI^B &= s \int_{y_{t,min}}^t \left(\frac{y_{t,min} - \mu}{s} - z \right) \phi(z) dz \\ &= \left((y_{t,min} - \mu) \Phi\left(\frac{y_{t,min} - \mu}{s}\right) + s \phi\left(\frac{y_{t,min} - \mu}{s}\right) \right) \\ &\quad - \left((y_{t,min} - \mu) \Phi\left(\frac{t - \mu}{s}\right) + s \phi\left(\frac{t - \mu}{s}\right) \right) \end{aligned} \quad (5)$$

$$\begin{aligned} EI^C &= -s \int_{2t - y_{t,min}}^t \left(\frac{2t - y_{t,min} - \mu}{s} - z \right) \phi(z) dz \\ &= - \left(((2t - y_{t,min} - \mu) \Phi\left(\frac{t - \mu}{s}\right) + s \phi\left(\frac{t - \mu}{s}\right)) \right. \\ &\quad \left. - ((2t - y_{t,min} - \mu) \Phi\left(\frac{2t - y_{t,min} - \mu}{s}\right) + s \phi\left(\frac{2t - y_{t,min} - \mu}{s}\right)) \right) \end{aligned} \quad (6)$$

The point given by $x = \arg \max(t - EI)$ is chosen for the next iteration. In the limit that $t \rightarrow \infty$, t-EI simplifies to EI³⁷ for finding maxima or minima, as shown in Fig. 2b.

Test cases for target-oriented BO method t-EGO

Sampling visualization. To visualize the dynamic optimization process of t-EGO and demonstrate how its performance differs from other methods, an investigation is conducted to search for a target value using a one-dimensional (1D) mathematical function (Supplementary Equation 1, Supplementary Fig. 1-3). For EGO and MOAF, the target value t , as well as the true and predicted curves, differ from those of the other methods due to the reformulation of the objective, as shown in Supplementary Fig. 1. With iterations, the evaluation score of t-EI converges faster compared to others, and the target is achieved with a difference of 0.0058(0.49%) within 4 iterations. Although the evaluation score of EI

gradually decreases, the objective-reformulated method EGO performs poorly and fails to reach the reformulated target t , ultimately falling into a local minimum in this case. The predictive distribution of the candidate and the integral interval in t-EI and EI, evolving with iterations, are further visualized in Supplementary Fig. 2. For EI, the integration interval is $[-\infty, y_{min}]$, which includes the infeasible region below t . This leads to an unreasonable estimation of the improvement interval, causing a low convergence rate. While for t-EI, the feasible region is reasonably controlled within $[t - y_{t,min}, y_{t,min}]$, and it gradually shrinks over iterations.

Synthetic function. To test the efficiency of the novel target-oriented BO method t-EGO compared with other existing BO methods, an investigation is conducted into searching for a target value involving the Styblinski-Tang function defined by $f(x) = \frac{1}{2} \sum_{i=1}^D (x_i^4 - 16x_i^2 + 5x_i)$, $x_i \in [-5, 5]$, for all $i = 1, \dots, D$ for feature dimension, $D = 2, \dots, 6$, where the target is -13.43 with error margin of $0.003 \times (y_{max} - y_{min})$. The function is shown for $D = 2$ in the 3D plot of Fig. 3a. In Fig. 3b-d, the optimization for $D = 6$ compares the performance of these five strategies over 100 repeated trials. Behavior up to $D = 5$ is shown in Supplementary Fig. 4.

An evaluation criterion opportunity cost(OC) illustrated in Methods is introduced to measure the efficiency of the BO methods and the iteration count needed in each trail is recorded. Fig. 3b shows OC for the five cases for dimension $D = 6$ as a function of the number of iterations over 100 trials. The OC curve for t-EGO rapidly decreases to converge to 0, whereas the target is not reached even after 100 iterations for the other strategies. On average, the target is attained using t-EGO in 14.19 additional new experiments, requiring approximately 1 to 2 times fewer experimental iterations than PureExp (30.7), EGO (39.17), MOAF (39.9) and CEGO (69.57) to reach the same target, as shown in Fig. 3c. The probability density function in Fig. 3d indicates that t-EGO needs 20 iterations or less to achieve the target in most cases (~79%). The long tails of the PDF curve for CEGO suggest that it performs poorly in approximately one-third of the trials. For PureExp, 49% of the 100 trials successfully located the target within 20 iterations, whereas 79% of the trials achieved the same outcome within 47 iterations. In comparison, only 40% and 35% of the trials achieve the target within 20 iterations for EGO and MOAF, with nearly 41% and 39% reaching the target between 30 and 70 iterations. Fig. 3e shows the closest distance from the target in the 100th iteration (OC_{100th}) as a function of dimensionality. EGO, MOAF and CEGO exhibit poorer performance in higher-dimensional settings. PureExp outperforms these methods, but its optimization efficiency declines as dimensionality increases. t-EGO demonstrates the capability to attain the target within 100 iterations across various dimensionalities, thereby exhibiting significant robustness.

The feasibility and robustness tests of the t-EGO strategy are further conducted on other four well-known high-dimensional standard mathematical test functions, in comparison with eight BO methods, including EGO, CEGO, MOAF, PureExp, Lower Confidence Boundary (LCB), Expected Quantile Improvement (EQI), Probability Improvement (PI) and Thompson Sampling (TS). The formulas for these benchmark functions, BO methods and the associated parameters are provided in Supplementary Equation 2-5 and Supplementary Tables 1-2. The results demonstrate that our method consistently outperforms others across test functions of varying complexity, achieving the target value with the fewest experimental iterations, as detailed in Supplementary Figs. 5-10 and summarized in Supplementary Fig. 11.

Active hydrogen evolution reaction catalyst. In the hydrogen evolution reaction, transition metals exhibit a volcanic distribution in the exchange current density corresponding to their hydrogen adsorption energy on the surface, with zero hydrogen adsorption energy ($\Delta G_H^* = 0$) at the center of the volcano³⁸. Transition metals with favorable catalytic activity are predominantly located at the center of the volcano, whereas transition metals with lower catalytic activity tend to be concentrated on both sides of the volcano³⁹⁻⁴².

Fig. 3 | Comparison of five strategies with different BO methods in search of a target value of -13.43 with error margin of $0.003 \times (y_{\max} - y_{\min})$ in 300 repeated trials. a Landscape of Styblinski-Tang function with dimensionality 2. **b** The opportunity cost with 95% confidence interval for five methods as the function of iterations. **c** The average iteration count and **d** PDF as the function of iteration count for each method in 100 trials. **e** The closest distance from the target in 100th iteration as the function of dimensionality.

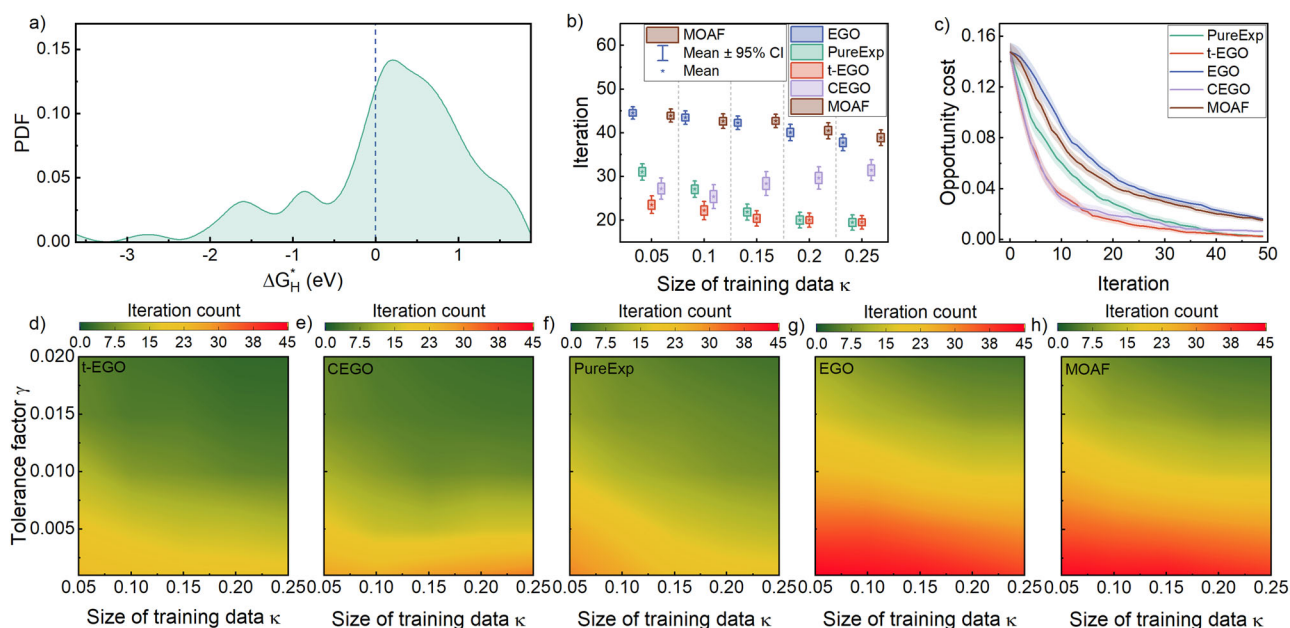
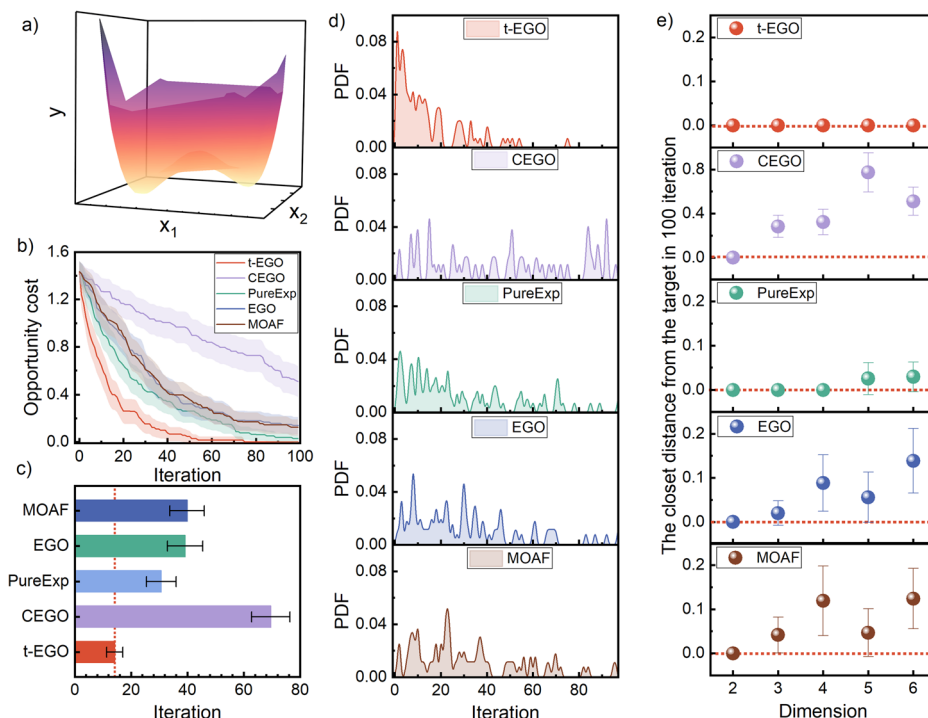


Fig. 4 | The results of searching for catalysts with $\Delta G_H^* = 0$ via five BO methods. a The distribution of ΔG_H^* in the catalytic database. **b** The efficiencies of BO methods change with different size of training data κ . Stars in the box plots stand by the

average number of iteration required. **c** Opportunity cost varies with iterations. **d–h** Heat maps show iteration count using t-EGO, CEGO, PureExp, EGO and MOAF as the function of training dataset size κ and tolerance factor γ .

Here we use a data set for 2D MA_2Z_4 containing 165 samples³⁰, where M is an early transition metal, A is C, Si, Ge, Sn, and Z is N, P, to screen catalysts with hydrogen adsorption free energy ΔG_H^* closest to zero. The distribution of ΔG_H^* in the catalytic database is plotted in Fig. 4a. Based on the feature pool, the four features: the weighted average boiling point (*Aver_W_Boi*), the average radius of the s orbital (*Aver_Ros*), maximum of the highest occupied molecular orbital Kohn-Sham eigenvalue (*Max_Homo*), the average periodic table group number (*Aver_Pc*), the minimum of the lowest unoccupied molecular orbital Kohn-Sham eigenvalue (*Min_Lumo*) are determined via feature engineering (Supplementary Fig. 12–13).

We investigate the dependence of five BO methods on the training set size κ , using sizes that are 0.05, 0.1, 0.15, 0.2, and 0.25 times the total population (165). Kriging models are established based on different training datasets for prediction. The functions t-EI together with EI, CEL, v_{PureExp} and v_{MOAF} are calculated to recommend the next candidate from the remaining data. The number of iterations that takes ΔG_H^* to satisfy $|\Delta G_H^*| \leq 0.003 \cdot (\Delta G_H^{*\max} - \Delta G_H^{*\min})$ is monitored over 200 trials for each training set, as shown in Fig. 4b. As the size of the training database increases, fewer iterations are required for these BO methods to reach the target, except for method CEGO. Overall, t-EGO performs the best, particularly when the

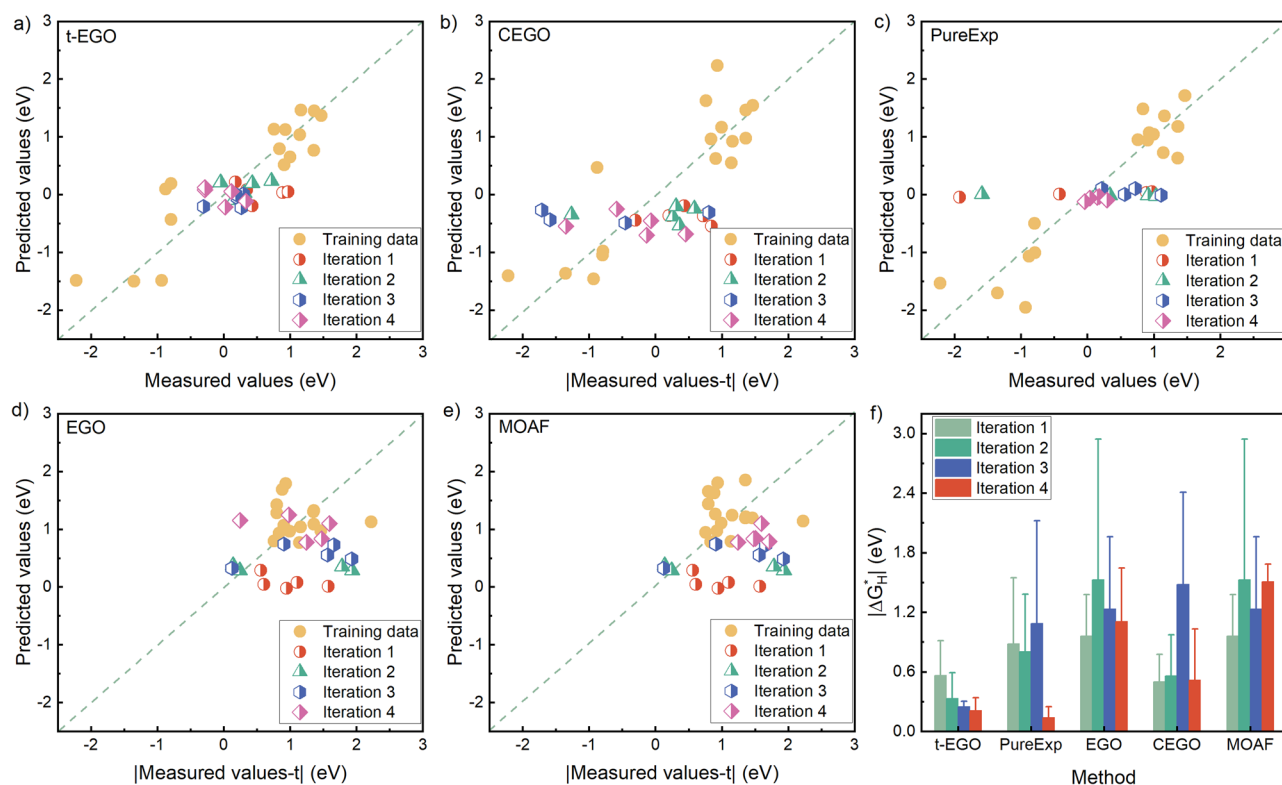


Fig. 5 | The results of catalyst searches with $\Delta G_H^* = 0$ using five BO methods, with five catalysts recommended in each iteration. a–c The measured values ΔG_H^{*mea} as a function of predicted values ΔG_H^{*pre} after 4 iterations for BO methods t-EGO, CEGO and PureExp, respectively. **d, e** The absolute value of the difference between the

measured value and the target value ($|\Delta G_H^{*mea} - t|$) as a function of predicted values after 4 iterations for BO methods EGO and MOAF. **f** The average value and standard deviation of $|\Delta G_H^*|$ for 5 candidates in each iteration.

training set size is small, such as when the size κ is 0.05 and 0.1 times the total population. Although the efficiency of PureExp improves as model accuracy increases with larger data sizes, t-EGO remains superior to PureExp. t-EGO requires approximately 1 to 2 times fewer iterations on average than EGO and MOAF, demonstrating a significant disparity even as the size of the training data increases. Figure 4c shows the OC as a function of iterations for $\kappa = 0.1$ in Fig. 4b. As the number of iterations increasing, OC gradually decreases to zero, which means the target is achieved. The OC decreases more rapidly for t-EGO, indicating that the ΔG_H^* of the selected samples by t-EGO approach the target $\Delta G_H^* = 0$ more quickly. Figure 4d–h shows the dependence of these BO methods on training dataset size κ and tolerance factor γ . A large training set size and a more relaxed tolerance result in fewer iterations to target. The t-EGO fares better than others, especially when $\gamma = 0.001$ and $\kappa = 0.05$, and on average, t-EGO requires 3 fewer, 7 fewer and 21 fewer iterations than CEGO, PureExp and EGO, respectively. Although CEGO requires a similarly low iteration count as t-EGO when $\gamma = 0.005/0.01$ and $\kappa = 0.1/0.15$, its iteration count increases as the size grows. Moreover, CEGO exhibits a disadvantage compared to t-EGO under stricter tolerance settings ($\gamma = 0.001$).

If 5 catalyst candidates are recommended each time for computational experiments, those selected by t-EGO tend to cluster around $\Delta G_H^* = 0$ (Fig. 5a). Except for two catalysts in the 1st iteration, all other recommended catalysts exhibit measured values that are closer to the target than those found in the initial dataset. Figure 5b exhibits a relatively scattered distribution of ΔG_H^* by using CEGO. Although fewer catalysts with ΔG_H^* near 0 are discovered after four iterations, the ΔG_H^* values of most catalysts are closer to 0 compared to the training data. The predicted values ΔG_H^{*pre} for candidates chosen through PureExp consistently hover near 0, yet the actual measured values, ΔG_H^{*mea} , deviate significantly, as shown in Fig. 5c. In contrast, for EGO and MOAF, the property values ΔG_H^{*mea} undergo pre-processing and are transformed into $y^{mea} = |\Delta G_H^{*mea} - t|$, where the actual values y^{mea} are strictly greater than 0, while some of the predictions y^{pre} are

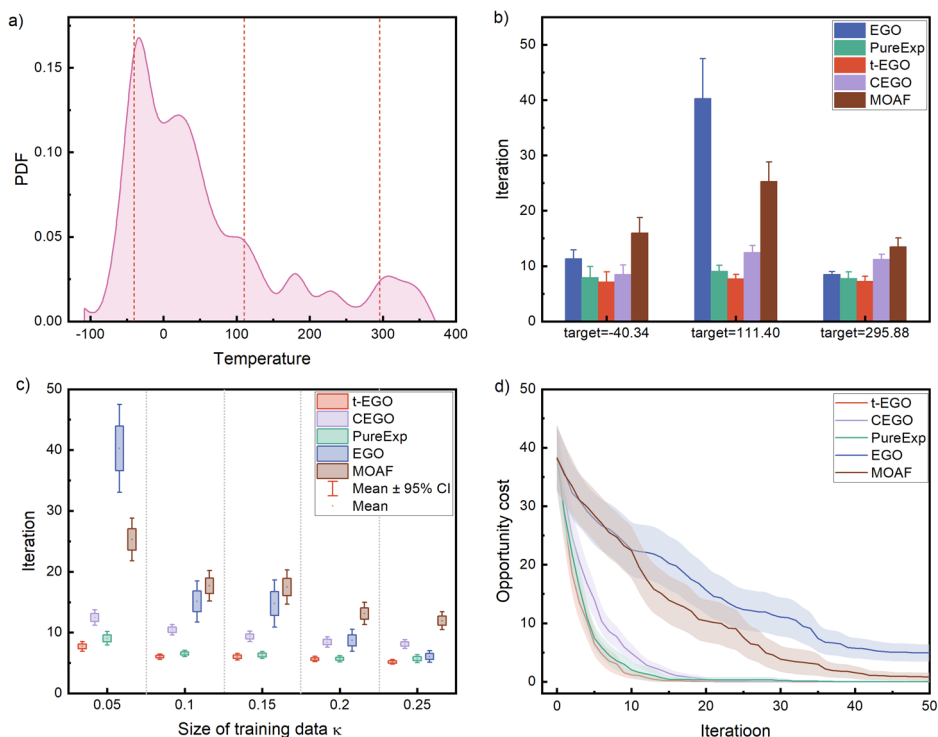
less than 0. Given this, applying the EGO and MOAF algorithm to minimize the value yields unreliable outcomes. Consequently, both the actual and predicted values exhibit a discrete distribution centered around 0 (Fig. 5d, e). The performance of the five methods is evaluated based on the average value of $|\Delta G_H^*|$ and their dispersion (Fig. 5f). In the 1st iteration, the $|\Delta G_H^*|$ values for the five candidates recommended via the t-EGO method exhibit a relatively low mean of 0.56 eV and a standard deviation of 0.35 eV. Moreover, as the number of iterations increases, the mean steadily decreases, reaching 0.21 with a standard deviation of 0.13 by the 4th iteration. In comparison, PureExp leads to a more scattered distribution of $|\Delta G_H^*|$ during the first three iterations, characterized by a mean greater than 0.8 eV and a standard deviation exceeding 0.5 eV. Until the fourth iteration, the predictions near the target value achieve relatively high accuracy following the addition of 15 data points to the database. EGO and MOAF exhibits the poorest performance in the four iterations, displaying the most scattered distribution of $|\Delta G_H^*|$ with a mean higher than 0.95 eV.

Search for shape memory alloys with target phase transformation temperature 440 °C. Shape memory alloys (SMAs), as thermally-responsive materials, are used to regulate valves, active devices, etc. at specific temperatures through the shape memory effect^{43,44}. In the present study, a high temperature shape memory alloy with a phase transformation temperature of 440 °C is desired for a thermostatic valve material for steam turbines to control the steam temperature. Doping in NiTi binary alloys can yield high temperature shape memory alloys. The extensive range of variations in elements and compositions poses challenges in identifying targeted SMAs.

We compare t-EGO to other BO methods and assess its performance on an SMA database before applying it to the experimental design of the desired SMAs. Data on 130 phase transformation temperatures of SMAs is gathered⁴¹, and the data distribution is depicted in Fig. 6a. The Kriging models are established using the database with different training sizes ($\kappa =$

Fig. 6 | The results of searching for SMAs with target temperature via five BO methods.

a Distribution of phase transformation temperature data for SMAs. **b** The average number of iterations and associated uncertainty for five BO methods in finding different target temperatures. **c** The efficiencies change with different size of training data. Stars in the box plots stand by the average number of iteration required. **d** When the size of training data κ is 0.05, the opportunity cost varies with iterations.



0.05, 0.1, 0.15, 0.2, 0.25) and materials descriptors previously derived including atomic number(anum), First ionisation energy(ion_energy), Young's Modulus(YM).¹¹ The target-specific properties are set to -40.34 °C, 111.40 °C and 295.88 °C, respectively, as indicated by the vertical red dashed lines in Fig. 6a. The average iteration results from 100 repeated trials starting from training datasets with size of $\kappa = 0.05$ for three different target properties are presented in Fig. 6b. The results demonstrate that t-EGO consistently required the fewest iterations across all target values. Figure 6c presents statistical results from 100 repeated trials with varying sizes of training datasets for identifying the phase transformation temperature of 111.40 °C. Although all BO methods perform better as the training data size increases, and the efficiency of PureExp approaches that of t-EGO at larger sizes, t-EGO exhibits superior optimization efficiency over other BO methods, particularly when the database size is very small (e.g. $\kappa = 0.05$), where its advantage becomes more pronounced. The OC curve of the dataset size $\kappa = 0.05$ is shown in Fig. 6d. We further investigate how the efficiency of these BO methods depends on the distribution of training data. An example is provided using a training dataset size of 0.05 times the total population to search for a SMA with a target property of 111.40 °C in the remaining data. Sorted by iteration count (N), we recorded the phase transformation temperature data of the training sets corresponding to the highest and lowest efficiency for each BO method in the 100 repeated trials, and visualized their distributions in Supplementary Figs. 14, 15. It is evident that t-EGO exhibits a lower sensitivity to the distribution of the training data. In contrast, the optimization efficiency of PureExp, EGO, MOAF, and CEGO strategies is influenced by the existing data points that are closest to the target. The performance of PureExp and CEGO also depends on the degree of data dispersion. Additionally, factors such as features and parameters in the model, which impact the accuracy of the model's predictions, may also serve as key determinants influencing the optimization efficiency.

The comparison of the five BO methods shows that t-EGO and PureExp methods are superior to others, so t-EGO and PureExp methods are employed to discover an SMA with a phase transformation temperature of 440 °C through experimental iterations. The candidate space of possible alloys is given by $(\text{Ti}_{u-x-y-\frac{z}{2}}\text{Zr}_x\text{Hf}_y)(\text{Ni}_{v-z-p-q-w-\frac{n}{2}}\text{Cu}_z\text{Fe}_p\text{Co}_q\text{Cr}_w)\text{Nb}_n$ alloys, where the compositions u, v, x, y, z, p, q, w , and n vary in specific

increments. The compositions are constrained by $0 \leq x \leq 0.1$, $0 \leq y \leq 0.24$, $0 \leq z \leq 0.14$, with step size 0.01, $0 \leq p, q \leq 0.07$, $0 \leq w \leq 0.03$, $0.48 \leq u, v \leq 0.52$ with a step size of 0.005 and $0 \leq n \leq 0.05$. It is also important to note that the compositions u and v are constrained by the condition $u + v = 1$. These constraints and step sizes define the range and granularity of the compositions within the candidate space of 279,946 possible potential alloys. The Kriging model is built using all 130 experimental data and the same material descriptors as described above. These 130 data in the training dataset represent about 0.046% of the 279,946 potential alloys in the unexplored space. The target phase transformation temperature of 440 °C differs by 83.56 °C from the closest temperature in the training database. The sampling process are visualized by projecting the candidates to a 2D space using t-Distributed Stochastic Neighbor Embedding(t-SNE) in Supplementary Fig. 16.

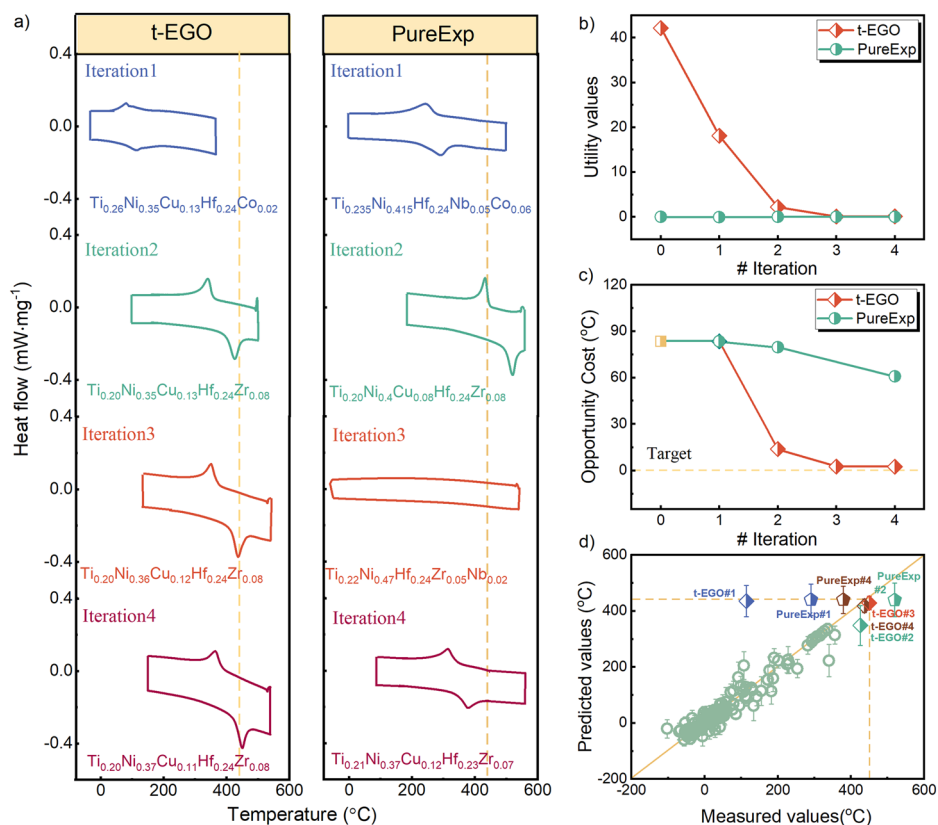
After 4 iterations, 8 alloys are synthesized and characterized using t-EGO and PureExp methods. The compositions and transformation temperatures of 8 newly synthesized alloys are listed in Table 1. One alloy does not undergo a phase transformation in the range of -60–580 °C. Figure 7a shows the DSC curves from differential scanning calorimetry for alloys using t-EGO and PureExp. The transformation temperature is determined from the peak value of the DSC curve observed during the heating process. Fig. 7a shows that the phase transformation temperature obtained in the 1st iteration is far from the target temperature, but then moves toward the target value in later iterations. The shape memory alloy $\text{Ti}_{0.20}\text{Ni}_{0.36}\text{Cu}_{0.12}\text{Hf}_{0.24}\text{Zr}_{0.08}$ guided using t-EGO in the 3rd iteration has a phase transformation temperature of 437.34 °C, which is very close to the target value of 440 °C, whereas those of the alloys suggested by PureExp after 4 iterations exhibit significant difference from the target temperature.

To determine whether to stop experimental iteration, the maximum utility values of all 279,946 alloys are recorded as iterations increase, as shown in Fig. 7b. The maximum utility given by t-EGO decreases from 42.16 in iteration #0 to 0.11 in iteration #3, which is closest to zero, showing that the experimental design can be stopped after three sampling iterations. We also monitor the OC in Fig. 7c to demonstrate the deviation of the current temperature from the target. The yellow half filled square in # 0 represent the minimum temperature difference (83.56 °C) between the

Table 1 | The alloy compositions recommended using different acquisition functions and the associated measured phase transformation temperatures

Strategy	Iteration	Alloys	Measured value /°C	Predicted value/°C
t-EGO	1	Ti _{0.26} Ni _{0.35} Cu _{0.13} Hf _{0.24} Co _{0.02}	114.34	435.87
	2	Ti _{0.20} Ni _{0.35} Cu _{0.13} Hf _{0.24} Zr _{0.08}	425.92	348.47
	3	Ti _{0.20} Ni _{0.36} Cu _{0.12} Hf _{0.24} Zr _{0.08}	437.34	418.57
	4	Ti _{0.20} Ni _{0.37} Cu _{0.11} Hf _{0.24} Zr _{0.08}	450.06	437.88
PureExp	1	Ti _{0.235} Ni _{0.415} Hf _{0.24} Nb _{0.05} Co _{0.06}	290.70	439.99
	2	Ti _{0.2} Ni _{0.40} Cu _{0.08} Hf _{0.24} Zr _{0.08}	519.70	439.93
	3	Ti _{0.22} Ni _{0.47} Hf _{0.24} Zr _{0.05} Nb _{0.02}	-60 ~ 580 °C no phase transformation	440.00
	4	Ti _{0.21} Ni _{0.37} Cu _{0.12} Hf _{0.23} Zr _{0.07}	379.10	440.00

Fig. 7 | Experimental results of discovering shape memory alloy with phase transformation temperature of 440 °C. a DSC curve of each alloy recommended using the acquisition functions t-EGO and PureExp, respectively. Noted that the top-ranked alloy evaluated by PureExp in the 3rd iteration does not undergo any phase transformation. The yellow vertical dotted line indicates the target temperature of 440 °C. **b** The maximum utility values of two acquisition function as the function of iterations. **c** Opportunity cost varies with iteration number. **d** Experimental measurements and predictions for alloys in training database and newly synthesized alloys are compared. The semi-filled prisms exhibit the results obtained by t-EGO. The horizontal dashed line in the figure represents the predicted temperature of the alloy as 440 °C, while the vertical dashed line represents the measured temperature of the alloy at 440 °C.



training data and the target. From the 2nd iteration, the experimentally measured phase transformation temperatures of the alloys progressively get closer to the target value compared to the origin training data. The OC curve of PureExp shows a relatively slow downward trend and remains 60.9 °C away from the target value even after 4 iterations, while t-EGO converges from 83.56 °C to 2.66 °C in the 3rd iteration. As shown in Fig. 7d, the predicted phase transformation temperatures for each iteration by PureExp are near the target 440 °C, but the corresponding measured values depart from it. This is because the lack of high-temperature data in the initial database can result in increasing uncertainties in the model predictions, with a significant deviation between the predicted and measured values. t-EGO exhibits a trend towards aligning more closely with the 45° diagonal line with increasing iterations, indicative of a progressively diminishing divergence between the predicted and measured values. The t-EGO algorithm demonstrates good robustness, capable of adjusting accordingly based on the prediction results, ensuring efficient completion of the design task regardless of the model's accuracy or high uncertainty.

Discussion

Focusing on the materials optimization problem of target-specific properties, we have designed a target-oriented Bayesian optimization method t-EGO to find a promising candidate with given target properties in an accelerated manner. This method incorporates the absolute difference between the predicted property values and the target-specific value into expected improvement, and allows the candidate properties to approach the target value from either direction, above or below, potentially reducing iteration count to identify target-specific properties. The optimization behavior of t-EGO is compared with the usual BO methods on mathematical functions of varying complexities, as well as two databases of catalysts and alloys. We further demonstrate its practical application experimentally through the synthesis and characterization of SMAs.

The t-EGO outperforms other BO methods, including EGO, CEGO, MOAF, and others, in identifying the target value within a given tolerance across the tested functions and material cases. The statistical results from 100 repeated trials show that the t-EGO strategy is 1 to 2 times more efficient

than the objective-reformulated strategies using EGO, MOAF, LCB, EQI, PI. This may benefit from reasonable constraint handling in t-EI, significantly improving the sampling efficiency. While in objective-reformulated strategies with EGO, MOAF, etc., the distribution of the predicted value is not adequately constrained. Integrating within the infeasible interval weakens the effectiveness of EI in querying the target value. In addition, LCB and EQI are also influenced by parameters in the formula. Although the CEGO strategy generally performs better than other objective-reformulated strategies in most scenarios, t-EGO still maintains superior efficiency across all cases. The performance of CEGO is inconsistent across different scenarios and appears to be influenced by the degree of dispersion in the training data around the target value. PureExp may achieve relatively rapid convergence in scenarios with large datasets, but its performance is limited in high-dimensional settings and cases with limited data availability. We have applied t-EGO in an experimental setting to discover thermally-responsive shape memory alloys with a target phase transformation temperature of 440 °C. The method leads to the synthesis of the shape memory alloy $\text{Ti}_{0.20}\text{Ni}_{0.36}\text{Cu}_{0.12}\text{Hf}_{0.24}\text{Zr}_{0.08}$ with a phase transformation temperature of 437.34 °C after only 3 new measurements, corresponding to a deviation of 2.66 °C from the target temperature. With the emerging interest and activity in autonomous self-driving laboratories for accelerated materials search, we believe that target-specific Bayesian optimization method t-EGO will readily be adopted where number of active learning iterations need to be minimized.

Methods

Workflow

Bayesian optimization loop with acquisition function for targeted property comprises the following: (1) We start with a training database of known compositions, materials descriptors and the corresponding properties. (2) We employ Kriging^{45,46} to build a surrogate model, screen unexplored materials and obtain predicted properties and associated uncertainties. The predicted mean and uncertainty are used in the next sampling step. (3) The acquisition function is used to rank all the materials according to their scores. Potential candidates with the highest scores are recommended for (4) simulations/synthesis and characterization. (5) Finally, the opportunity cost (OC) is calculated. If OC is smaller than a predefined threshold, the iteration stops. Otherwise, the newly measured experimental data augments the training database, and the active learning loop is repeated.

Modeling

Training data set for validation is selected by subsampling techniques⁶. We construct training data set by randomly drawing n data from the original total database without replacement, and n usually is a portion of the total quantity (eg. 30%). The Kriging model served as a machine learning model is implemented by DiceKriging package in the statistical studio R provided by Roustant et al.⁴⁷. Hyperparameters such as length scale, variance, and others are optimized using Maximum Likelihood Estimation (MLE), with the BFGS method as the optimization algorithm. This model⁴⁵ was employed for gaining the predicted distributions of the unexplored candidates, including the prediction means as well as the estimated uncertainties for each prediction. Leave-One-Out Cross-Validation was used and combined with Kriging to evaluate model performance.

Evaluation criterion

This evaluates the efficiency of the acquisition function and determine whether to stop experimental iteration. OC calculates the minimum distance between the target value t and the measurement values obtained so far. That is,

$$OC = \min(|t - y|, |t - y_{t,min}|) \quad (7)$$

where y presents the measured value of the new sample; $y_{t,min}$ represents the measurement value in the training set that is closest to the target value. The iterations can be terminated in terms of the OC value approaching 0, which indicates that the measurement value of the new sample is close to the target value and meets the requirements. On the other hand, the fewer the number of iterations required for the OC value to reach the stopping threshold, the more efficient the Bayesian optimization method. Thus, a smaller OC indicates higher efficiency of the Bayesian optimization method.

Experimental methods

The as-cast ingots of the SMAs are prepared by arc melting from pure Ti (99.995%), Ni (99.995%), Hf (99.99%), Zr (99.99%), Co (99.98%), Cu (99.99%), and Nb (99.5%) in an argon atmosphere. The specimens for measurements are spark-cut with the size of 3 mm × 3 mm × 1 mm from the ingots. Then they are solution-treated at 1273 K for 1 h in evacuated quartz tubes, followed by water quenching. The DSC curves are obtained via Netzsch 214 differential scanning calorimetry (DSC) measurements with a heating/cooling rate of 10 K · min⁻¹. The inverse martensitic transformation temperatures are determined using endothermic peak values during heating.

Data availability

The data that support the findings of this study are available from https://github.com/tiany0321/target_oriented_BO and https://github.com/diegouxue/target_oriented_BO.

Code availability

The codes that support the findings of this study are available from https://github.com/tiany0321/target_oriented_BO and https://github.com/diegouxue/target_oriented_BO.

Received: 11 December 2024; Accepted: 13 June 2025;

Published online: 03 July 2025

References

- Lookman, T., Balachandran, P. V., Xue, D. & Yuan, R. Active learning in materials science with emphasis on adaptive sampling using uncertainties for targeted design. *npj Comput. Mater.* **5** (2019).
- Tian, Y. et al. Noise-aware active learning to develop high-temperature shape memory alloys with large latent heat. *Adv. Sci.* **11**, 2406216 (2024).
- Xue, P. et al. Solutions for lithium battery materials data issues in machine learning: Overview and future outlook. *Adv. Sci.* **11**, 2410065 (2024).
- Yang, Z. & Gao, W. Applications of machine learning in alloy catalysts: Rational selection and future development of descriptors. *Adv. Sci.* **9**, 2106043 (2022).
- Wahl, C. B. et al. Machine learning-accelerated design and synthesis of polyelemental heterostructures. *Sci. Adv.* **7** (2021).
- Musil, F., Willatt, M. J., Langovoy, M. A. & Ceriotti, M. Fast and accurate uncertainty estimation in chemical machine learning. *J. Chem. Theory Comput.* **15**, 906–915 (2019).
- Kim, M. et al. Searching for an optimal multi-metallic alloy catalyst by active learning combined with experiments. *Adv. Mater.* **34**, 2108900 (2022).
- Zhang, H., Chen, W., Iyer, A., Apley, D. W. & Chen, W. Uncertainty-aware mixed-variable machine learning for materials design. *Sci. Rep.* **12** (2022).
- Vandermause, J. et al. On-the-fly active learning of interpretable bayesian force fields for atomistic rare events. *npj Comput. Mater.* **6** (2020).
- Kusne, A. G. et al. On-the-fly closed-loop materials discovery via bayesian active learning. *Nat. Commun.* **11** (2020).
- Tian, Y. et al. Determining multi-component phase diagrams with desired characteristics using active learning. *Adv. Sci.* **8**, 2003165 (2021).

12. Tian, Y. et al. Efficient estimation of material property curves and surfaces via active learning. *Phys. Rev. Mater.* **5**, 013802 (2021).
13. Saleh, Y., Sanjay, V., Iske, A., Yachmenev, A. & Küpper, J. Active learning of potential-energy surfaces of weakly bound complexes with regression-tree ensembles. *J. Chem. Phys.* **155**, 144109 (2021).
14. Friederich, P., Häse, F., Proppe, J. & Aspuru-Guzik, A. Machine-learned potentials for next-generation matter simulations. *Nat. Mater.* **20**, 750–761 (2021).
15. Bull, A. D. Convergence rates of efficient global optimization algorithms. *J. Mach. Learn. Res.* **12**, 2879–2904 (2011).
16. Srinivas, N., Krause, A., Kakade, S. M. & Seeger, M. Gaussian process optimization in the bandit setting: No regret and experimental design. *Proceedings of the 27th International Conference on International Conference on Machine Learning, ICML '10, 1015–1022* (Omnipress, Madison, WI, USA, 2010).
17. Srinivas, N., Krause, A., Kakade, S. M. & Seeger, M. W. Information-theoretic regret bounds for Gaussian process optimization in the bandit setting. *IEEE Trans. Inf. Theory* **58**, 3250–3265 (2012).
18. Shields, B. J. et al. Bayesian reaction optimization as a tool for chemical synthesis. *Nature* **590**, 89–96 (2021).
19. Attia, P. M. et al. Closed-loop optimization of fast-charging protocols for batteries with machine learning. *Nature* **578**, 397–402 (2020).
20. Tian, Y., Lookman, T. & Xue, D. Efficient sampling for decision making in materials discovery*. *Chin. Phys. B* **30**, 050705 (2021).
21. Xue, D. et al. Accelerated search for materials with targeted properties by adaptive design. *Nat. Commun.* **7** (2016).
22. Yuan, R. et al. Accelerated discovery of large electrostrains in batio3-based piezoelectrics using active learning. *Adv. Mater.* **30**, 1702884 (2018).
23. Wen, C. et al. Machine learning assisted design of high entropy alloys with desired property. *Acta Mater.* **170**, 109–117 (2019).
24. Wen, D., Tucker, V. & Titus, M. S. Bayesian optimization acquisition functions for accelerated search of cluster expansion convex hull of multi-component alloys. *npj Comput. Mater.* **10**, 210 (2024).
25. Vasylenko, A. et al. Inferring energy-composition relationships with bayesian optimization enhances exploration of inorganic materials. *J. Chem. Phys.* **160**, 054110 (2024).
26. Cowen-Rivers, A. I. et al. Hebo: Pushing the limits of sample-efficient hyperparameter optimisation. *J. ARTIF. INTELL. RES.* **74**, 1269–1349 (2022).
27. Yin, P. et al. Machine-learning-accelerated design of high-performance platinum intermetallic nanoparticle fuel cell catalysts. *Nat. Commun.* **15** (2024).
28. Geetha, M., Singh, A., Asokamani, R. & Gogia, A. Ti based biomaterials, the ultimate choice for orthopaedic implants - a review. *Prog. Mater. Sci.* **54**, 397–425 (2009).
29. Niinomi, M. & Nakai, M. Titanium-based biomaterials for preventing stress shielding between implant devices and bone. *Int. j. biomater.* 836587 (2011).
30. Zheng, J. et al. Symbolic transformer accelerating machine learning screening of hydrogen and deuterium evolution reaction catalysts in ma2z4 materials. *ACS Appl. Mater. & Interfaces.* **13**, 50878–50891 (2021).
31. Xiao, Z., Meng, W., Wang, J., Mitzi, D. B. & Yan, Y. Searching for promising new perovskite-based photovoltaic absorbers: the importance of electronic dimensionality. *Mater. Horiz.* **4**, 206–216 (2017).
32. Ma, X.-Y. et al. High-efficient ab initio bayesian active learning method and applications in prediction of two-dimensional functional materials. *Nanoscale* **13**, 14694–14704 (2021).
33. Cui, J. et al. Combinatorial search of thermoelastic shape-memory alloys with extremely small hysteresis width. *Nat. Mater.* **5**, 286–290 (2006).
34. Chitturi, S. R. et al. Targeted materials discovery using bayesian algorithm execution. *npj Comput. Mater.* **10** (2024).
35. Schonlau, M., Welch, W. J. & Jones, D. R. Global versus local search in constrained optimization of computer models. *Lecture notes-monograph series* 11–25 (1998).
36. Jiao, R., Zeng, S., Li, C., Jiang, Y. & Jin, Y. A complete expected improvement criterion for gaussian process assisted highly constrained expensive optimization. *Inf. Sci.* **471**, 80–96 (2019).
37. Jones, D. R., Schonlau, M. & Welch, W. J. Efficient global optimization of expensive black-box functions. *J. Glob. Optim.* **13**, 455–492 (1998).
38. Ekspong, J., Gracia-Espino, E. & Wågberg, T. Hydrogen evolution reaction activity of heterogeneous materials: a theoretical model. *J. Phys. Chem. C.* **124**, 20911–20921 (2020).
39. Ahmed, M., Kour, G., Sun, Z., Du, A. & Mao, X. Activating hydrogen evolution reaction on carbon nanotube via aryl functionalisation: The role of hybrid sp2–sp3 interface and curvature. *Nanomaterials* **13** (2023).
40. Nørskov, J. K. et al. Trends in the exchange current for hydrogen evolution. *J. Electrochem. Soc.* **152**, J23 (2005).
41. Quaino, P., Juarez, F., Santos, E. & Schmickler, W. Volcano plots in hydrogen electrocatalysis - uses and abuses. *Beilstein J. Nanotechnol.* **5**, 846–854 (2014).
42. Li, T., Yu, Y. & Pei, M. Dual-atom doping carbon materials as highly efficient electrocatalysts for lithium-sulfur batteries: Bimetallic cooperation mechanism. *J. Phys. Chem. C.* **127**, 6271–6279 (2023).
43. Hart, G. L. W., Mueller, T., Toher, C. & Curtarolo, S. Machine learning for alloys. *Nat. Rev. Mater.* **6**, 730–755 (2021).
44. McCracken, J. M., Donovan, B. R. & White, T. J. Materials as machines. *Adv. Mater.* **32**, 1906564 (2020).
45. CRESSIE, N. The origins of kriging. *Math. Geol.* **22**, 239–252 (1990).
46. Ginsbourger, D., Le Riche, R. & Carraro, L. Kriging Is Well-Suited to Parallelize Optimization, 131–162 (Springer Berlin Heidelberg, Berlin, Heidelberg, 2010).
47. Roustant, O., Ginsbourger, D. & Deville, Y. Dicekriging, diceoptim: Two r packages for the analysis of computer experiments by kriging-based metamodeling and optimization. *J. Stat. Softw.* **51**, 1–55 (2012).

Acknowledgements

We gratefully acknowledge the support of the National Key Research and Development Program of China (2021YFB3802102), National Natural Science Foundation of China (Grant Nos. 52303297, 52350710205), and China Postdoctoral Science Foundation (2022TQ0201, 2023M742196).

Author contributions

The study was planned and designed by T. Lookman, X.D. and D.X., machine learning programs and experiments were performed by Y.T. and J.P., catalysts data collection and analysis were done by Y.T. and T.Li, the manuscript was prepared by Y.T., Y.Z., D.X., and T. Lookman. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41524-025-01704-4>.

Correspondence and requests for materials should be addressed to Dezhen Xue, Xiangdong Ding or Turab Lookman.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025